D1A — Research Questions, Materials and Protocol

Appliance Manual Usability — Empathize Phase Research Plan

Course	CS3014 — Applied Human Computer Interaction
Deliverable	D1A — Research Questions, Materials and Protocol
Team	*Konnect*
Team Members	*23i-0809,23i-0504,23i-0581,23i-0786*
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1. Research Question

D0 Assumption Traced To:

> *"Users struggle to set up or troubleshoot household appliances because physical user manuals fail to visually map components to their practical spatial locations and functions."*

D1A Research Question (Behavior-Focused):

> *How do adult householders currently locate, identify, and understand specific appliance components and operational steps when attempting maintenance or setup tasks using physical or digital instruction manuals?*

This question targets current behavior and mental-model breakdowns around spatial orientation and terminology — it does not ask whether participants want an app, and no proposed solution is named or implied anywhere in the question.

2. Participant Criteria & Sampling Strategy

Inclusion Criteria	Exclusion Criteria
Adults aged 18–60 who personally assemble, maintain, or troubleshoot everyday household appliances (e.g., espresso machines, vacuum cleaners, food processors) in their own home. Have attempted to follow a physical or digital manual within the past 6 months.	Professional technicians or appliance repair experts (expertise removes the novice confusion the study targets). Vulnerable populations — no minors, patients, or other groups requiring formal ethical review.

Sampling Strategy

Purposive sampling.

We will deliberately select participants who own and regularly maintain multi-part household appliances, rather than sampling at random or by convenience alone. This is justified because the research question requires participants who have recent, concrete manual-use experiences to draw on during the observation task and interview — a random or purely convenience-based sample (e.g., "whoever is available") risks recruiting people with little or no relevant recent experience, producing thin or hypothetical answers. Purposive sampling is also feasible within the team's fieldwork window, since appliance ownership and manual use are common enough that criteria-matching participants can be identified within our existing networks without needing snowball referrals.

3. Consent Materials & Data-Management Plan

Consent Script (read aloud, verbatim, before every session)

> “Thank you for agreeing to speak with us today. This session is part of a university HCI research project on how people currently locate, identify, and understand appliance components and instructions when using manuals for maintenance or setup tasks. There are no right or wrong answers — we're interested in what you actually do.This will take about 20 minutes. With your permission, we'd like to record audio of the session so we don't have to take notes the whole time — you're welcome to decline, and we'll take written notes instead. Everything you say will be anonymized before it appears in anything we write up — you'll be referred to as, for example, 'Participant A' — and only our project team will have access to the raw recording. Recordings and notes will be stored in a password-protected [local folder / university-approved cloud drive] and permanently deleted at the end of the academic term. No identifiable data or raw recordings will be uploaded to any external AI tool.Your participation is completely voluntary. You can skip any question or stop the session at any point, for any reason, without needing to explain why and without any penalty. Do you have any questions before we begin? Are you comfortable proceeding?”

This script covers, per the D1A requirements: the purpose of the study, expected duration, what data is collected and how, who sees it and how it is stored, that participation is voluntary, and that the participant can stop at any time.

Data-Management Plan

- Storage: raw audio and notes stored only in a password-protected [local team folder / university-approved cloud drive] — never on personal, unsecured devices.
- Access: only current team members have access to raw recordings and notes; no external collaborators or tools.
- Anonymization: transcripts and reports refer to participants only by session label (e.g., Participant A, Participant B) — no names, contact details, or identifying images are retained in any shared document.
- External AI tools: no identifiable participant data or raw audio/video is uploaded to any external AI tool at any stage; only anonymized text is used, if any AI-assisted analysis is applied later in the project.
- Retention: all raw data (recordings, notes, consent records) is permanently deleted at the end of the academic term [insert specific date]; anonymized summary findings may be retained for the remainder of the course.
- Withdrawal: a participant may request their data be removed at any point up to [insert cutoff date] by contacting the team at [team contact email].

4. Piloted Protocol

Logistics

- Duration: ~20 minutes
- Format: one-on-one contextual inquiry / observation task + short interview, in person
- Materials: this protocol, consent script (above), an unfamiliar multi-part appliance with its manual, audio recorder (with permission), notebook

Opening / Consent (read aloud)

Identical to the Consent Script above (Section 3) — reproduced here so the interviewer can run the session from a single document.

> “Thank you for agreeing to speak with us today. This session is part of a university HCI research project on how people currently locate, identify, and understand appliance components and instructions when using manuals for maintenance or setup tasks. There are no right or wrong answers — we're interested in what you actually do.This will take about 20 minutes. With your permission, we'd like to record audio of the session so we don't have to take notes the whole time — you're welcome to decline, and we'll take written notes instead. Everything you say will be anonymized before it appears

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Warm-up

1. Tell me about the last appliance you bought or had to fix or clean. Did you open the manual? What happened?

Core Observation & Questions

2. Here is a manual for airfrier. Please show me how you would find where the [part name / filter] is located and how to remove it. (Observe: where do their eyes go on the page? Where do they hesitate or flip back and forth?)
3. When a step refers to a specific named part, how do you verify you are holding or adjusting the right piece?
4. Can you walk me through a time when a manual's instructions failed to help you complete a task?

Planned Probes

- "What made that specific diagram or text confusing?"
- "Where did you look next when that step didn't make sense?"
- "Can you say more about that?" (general follow-up if an answer is thin)

Closing (read aloud)

"That's everything we wanted to cover — thank you so much, this was really helpful. If you think of anything else afterward, or you'd like any part of what you said removed, you can reach us at [team contact email] any time before [insert cutoff date]. Thanks again for your time."

Pilot Note (Required Component)

> *"We piloted this protocol with a student using an airfrier manual. Our initial protocol asked, 'Do you think an app with 3D/visual steps would make this manual easier?' The participant immediately pitched app features instead of attempting the task, making the session solution-oriented. We revised the protocol to present a physical manual and asked them to complete a task while thinking out loud, focusing purely on where and why they got stuck. No second pilot was needed — the revised version ran smoothly end to end when re-read through with the same participant."*

Before / After Summary

Draft version (pre-pilot)	Issue found during pilot	Revised version (piloted)
"Do you think an app with 3D/visual steps would make this manual easier?"	Leading and solution-oriented — the pilot participant immediately began pitching app features instead of attempting the task.	"Here is a manual for airfrier. Please show me how you would find where the [part/filter] is and how to remove it." (task-based, observed)

5. Hard Ethical Constraints — Confirmed

- No vulnerable populations (minors, patients, or others requiring formal ethical review) are included in recruitment or the exclusion criteria above.

- No identifiable participant data or raw audio/video will be uploaded to any external AI tool at any stage of analysis.